Detecting Exoplanet Transits with the Keeble Telescope

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Introduction

Objective: To confirm the presence of large exoplanets around M-dwarf stars in collaboration with GEMS (Giant Exoplanets around M-dwarf Stars). This is a follow-up project with planetary candidates using test data from TESS.

Brief Conceptual Overview: An exoplanet is a planet that resides outside of our solar system. These exoplanets have transits, which occur under the circumstances of the planet passing between the star and the observer.

Many exoplanets have been discovered through the *transit method*; this method utilizes photometry by recording the changes in flux from the exoplanet's host star and then graphing the detected values into a light curve.

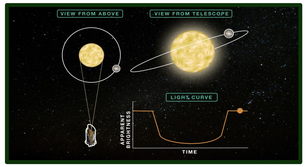


Figure 1: Example of exoplanet passing between host star and observer
Credit: viewspace.org

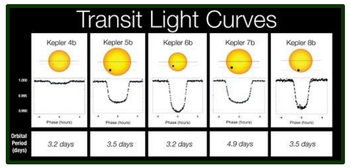


Figure 2: Example of light curves from various exoplanets
Credit: NASA

Observatory

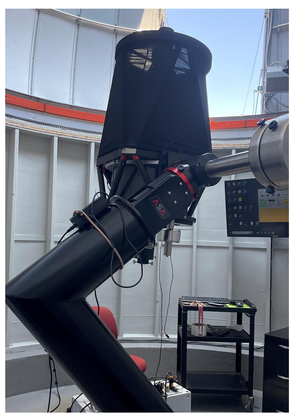


Figure 3: Telescope in Keeble Observatory

About Keeble: Although Keeble observatory's existence is over 60 years old, it was recently rebuilt in 2017. Inside features the telescope, an ASA 40 cm Ritchey-Chretien F/8.The dome is controlled by ACE SmartDome and ASCOM POTH hub, meanwhile the telescope relies on Autoslew and Cartes du Ciel for movements and ACC for shutter and focal adjustments.

About the CCD: The CCD, or Charge-Coupled Device, is the camera used to capture images of the night sky. It is a SBIG ST-10XMEI with thermoelectric cooling and a pixel resolution of 0.88 arcsec per pixel. The camera operates through Maxim DL. We rely on an infrared filter (Kron/Cousins I-Band) to minimize effects from light pollution. Therefore, the filter only allows infrared light to pass through.

Data and Results

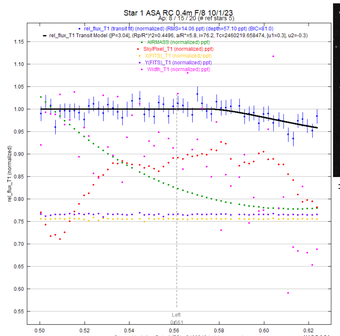


Figure 4: Light curve for Star 1 transit

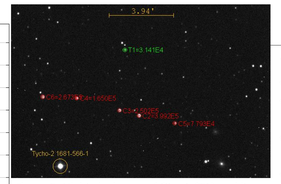


Figure 5: Night sky image of Star 1 (T1) and its comparison stars (C2 to C6)

Star 1

Host star:
- 14.5 mag. (*TESS*)
- Temperature: 3413± 157 K
- Mass: 0.463± 0.021 $M_{\odot}$
- Radius: 0.465± 0.015 $R_{\odot}$

Exoplanet:
- Radius: 3.06 R_{jup}
- Date: 10/01/23
- Depth: 57.10 ppt
- Transit (UTC): 01:28—01:56—02:23

Data: The total time of the observing run is approximately 3 hours. All science image exposures are 180 seconds long at -9˚ C. 20 exposures were taken in the span of an hour, totaling to 57 images overall.

Results: As shown in Figure 4, the data captures the ingress of the transit, but the egress is not present. From the data, a value for the planetary radius and depth of transit could be determined. The calculated planetary radius is 3.06 R_{jup} and the depth of transit is 57.10 ppt. There is uncertainty for the transit center value, since the entirety of curve is not available.

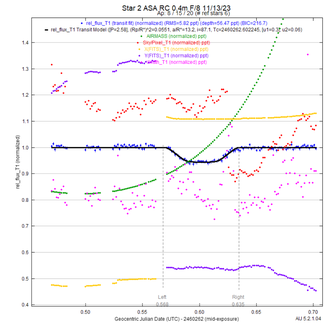


Figure 6: Light curve for Star 2 transit

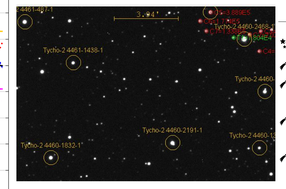


Figure 7: Night sky image of Star 2 (T1) and its comparison stars (C2 to C7)

Star 2

Host star:
- 13.2 mag. (*TESS*)
- Temperature: 3628± 157 K
- Mass: 0.496± 0.0203 $M_{\odot}$
- Radius: 0.498± 0.0149054 $R_{\odot}$

Exoplanet:
- Radius: 1.15 R_{jup}
- Date: 11/13/23
- Depth: 56.47 ppt
- Transit (UTC): 01:38—02:26—03:14
- T_c (UTC): 2:27

Data: The total time of the observing run is approximately 5 hours. All science image exposures are 120 seconds at -17˚ C. 30 exposures were taken in the span of an hour, totaling to 138 images.

Results: A value for the planetary radius, depth of transit, and transit center was calculated. The data provided a planetary radius of. The depth of the transit is 56.47 ppt and the transit center is 2:27 UTC.

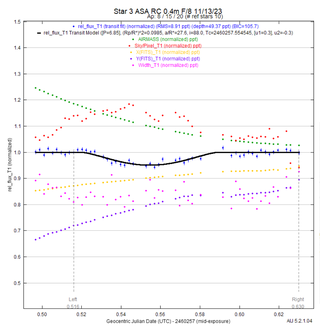


Figure 8: Light curve for Star 3 transit

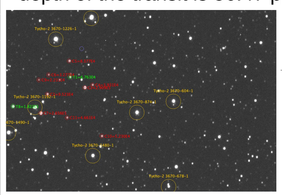


Figure 9: Night sky image of Star 3 (T1) and its comparison stars (C2 to C9)

Star 3

Host star:
- 13.8 mag. (*TESS*)
- Temperature: 3522± 157 K
- Mass: 0.473± 0.021 $M_{\odot}$
- Radius: 0.475± 0.014 $R_{\odot}$

Exoplanet:
- Radius: 0.86 R_{jup}
- Date: 11/08/23
- Depth: 49.37 ppt
- Transit (UTC): 00:23—01:09—01:54
- T_c (UTC): 1:18

Data: The total time of the observing run is approximately 4 hours. All science image exposures are 180 seconds at -17˚ C. 20 exposures were taken in the span of an hour, totaling to 79 images.

Results: From the data, a value for the planetary radius, depth of transit, and transit center was provided. A planetary radius value of 0.86 R_{jup} is provided. For the depth of the transit, the value is 49.37 ppt and the transit center is 1:18 UTC.

Methods

Observations: Nights of observations typically begin prior to sunset to take calibration images. The transit data is taken in several sets in the duration of one night. Many of the dark fields were taken independent of the observing nights.

Analysis: The data is reduced through the CCD Data Processor in AstroImageJ. The reduced, new images are then imported into AstroImageJ and plate-solved. A light curve is produced via differential photometry.

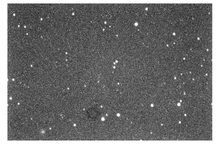


Figure 11: *Star 2*, before plate solve and reduction (*10/01/23*), 180 second exposure

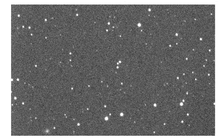


Figure 12: *Star 2*, after plate solve and reduction

Conclusions & Future

Conclusion: We present the findings of three plausible exoplanet transits around M-dwarf stars. These exoplanets are of Jupiter-size with large transit depths and have a host star with a faint magnitude. In these findings, Keeble telescope detected exoplanet transits with planetary radius values that range between 0.86 and 3.06 R_{jup} .

Future Work: Some considerations for future research are to investigate some of the common issues that interfered with the project. The main offender was subpar telescope tracking, which ruined many of the images. One of the possible solutions may be to adjust the balance of the telescope. This project has been a collaboration with GEMS and is intended to continue in the future.

References

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